SLoMO Reproducibility Documentation

Abstract. This report describes an auditable movie-level pipeline for open-ended long-video question answering. All questions for one film share a timestamped transcript and chronological visual observations, and one validated JSON answer bank is produced. The release contains code, chart data, tests, and implementation notes; datasets, weights, predictions, credentials, and evaluation records remain external inputs.

Keywords: long-video QA · movie-level inference · reproducibility

1 System Overview

The movie is the context unit. A shared request lets the answerer preserve aliases, relationships, event order, causes, and ending state across questions. The release contains inference and submission utilities, tests, configuration guidance, and evidence-bound diagnostic plots. Videos, subtitles, weights, labels, generated answers, accounts, and machine paths are excluded.

2 Method

For movie m, let E_m be shared evidence and Q_m its complete question set. The runner computes

$$(E_m, Q_m) \longrightarrow \{a_{m,1}, a_{m,2}, \ldots, a_{m,n_m}\}.$$

It groups rows by `video_id`, parses timestamped subtitle cues, samples 16 chronological frames, and sends one request containing all movie questions. The model returns one JSON object keyed by the original IDs. The prompt requests concise English answers, cross-question consistency, and evidence support. Frames are JPEG encoded at a fixed maximum side and marked with timestamps; long transcripts use a head-and-tail bound.

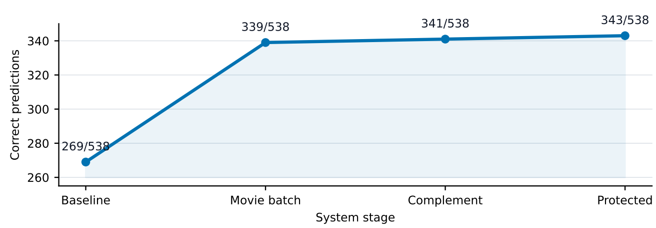

Figure 1. Historical official score chain. The large discontinuity coincides with a movie-batched multimodal answer source; later parent-preserving changes are smaller.

3 Inference Pipeline

The pipeline groups annotations, parses timestamped subtitles, samples the chronological frame pack, sends one movie-level request, and validates the complete JSON bank before persistence. Transient provider failures may be retried. Missing IDs, empty answers, malformed JSON, and partial movie banks are errors and are never silently merged.

4 Reproducibility

The runner accepts an OpenAI-compatible Responses endpoint and reads its key only from a process environment variable. The conversion utility writes the required `question_id,prediction` CSV. The audit utility checks the header, exact ID coverage, uniqueness, and non-empty predictions. Public and private splits are supplied independently.

Component	Configuration
Context	One movie; all questions jointly
Evidence	Timestamped transcript + 16 chronological JPEG frames
Frame encoding	Maximum side 512 px; JPEG quality 82
Generation	Temperature 0; high reasoning effort
Output	JSON bank, then `question_id,prediction` CSV

5 Results and Ablations

Figure 2 reports a frozen six-movie, 60-question local diagnostic using one semantic evaluator. Joint movie answering reaches 27/60, independent calls 18/60, and no frames 24/60. Target16 and Dense48 reach 29/60 and 30/60. The result supports joint context; the controlled visual changes do not establish a large standalone gain.

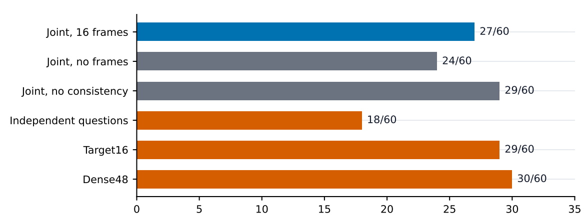

Figure 2. Frozen factor and visual-ceiling diagnostic on six movies. Blue is the direct reference; gray and orange bars are controlled variants. These are local diagnostics, not official leaderboard scores.

Limitations. Performance depends on transcript quality, visual coverage, endpoint behavior, and context budget. Data and weights are external inputs; reproduction requires authorized access and the included schema audit. Public sources: SLoMO, EvalAI, and SF20K.